# Infrastructure Defense Demo

Microsoft Defender for Endpoint and IoT

Based on the Aliquippa Cyberattack

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# Microsoft Defender for Endpoint and Defender for IoT to Protect Small Utility Systems

In 2024, more than 70% of inspected water utilities in the U.S. were found not to meet the Safe Drinking Water Act's cybersecurity requirements (Hearst Television, 2024). The cyber-attack on the Municipal Water Authority of Aliquippa (MWAA) is one of the examples, showing a critical security vulnerability of small utility systems. In November 2023, a pro-Iranian hacktivist group known as cyberAv3ngers hacked a publicly accessible Programmable Logic Controller (PLC) used at a water booster station of MWAA. This group used default credentials to access and disable the PLC to leave a politically charged defacement message saying, "YOU HAVE BEEN HACKED. DOWN WITH ISRAEL". While there weren't other known physical impacts on the system, the attack underscored how easily infrastructure can be compromised when not properly secured. This demo reports how implementing Microsoft Defender for Endpoint and Microsoft Defender for IoT as a scalable and integrated solution secures small-to-medium utility systems like MWAA.

## Business Case and Need

Recently, the rate of attacks on Municipal utilities targeted by state-sponsored cyberattacks has significantly increased due to a lack of full-time IT security teams and reliance on inefficient infrastructure. In the case of Aliquippa, the PLC used outdated security practices and lacked basic protections such as encryption or authentication. Microsoft Defender for Endpoint provides real-time threat detection and automated response, while Defender for IoT extends monitoring to industrial control systems such as PLCs. Together, they offer a unified approach that can alert, isolate, and mitigate attacks across both IT and operational technology (OT) systems, resolving the primary issue of the case.

This also aligns with the security governance and risk management principle from Chapter 1 (Chapple, Stewart, & Gibson, 2024), emphasizing the understanding of the business context, conducting risk assessments, and applying controls to protect assets in alignment with organizational objectives. Additionally, the need for layered defense also reflects the security and risk management concepts of Chapter 2, particularly around preventive, detective, and corrective control implementation. And lastly, this aligns with Chapter 3 on security architecture and engineering, the defense-in-depth principle, which advocates for multi-layered protection (Chapple, Stewart, & Gibson, 2024).

**Table 1:** *Side-by-Side Tool Comparison*

| Tool | Protects IT (Laptops/Servers) | Protects OT (PLCs/SCADA) | Active Protection | Passive Monitoring |
|---|---|---|---|---|
| **Defender for Endpoint** | Yes | No | Yes | No |
| **Defender for IoT** | No | Yes | No | Yes |

**Cost and Benefit Comparison**

In comparison to its competitors, Microsoft Defender for Endpoint offers great integration for organizations already utilizing Windows. Even though CrowdStrike Falcon and SentinelOne offer great detection capabilities, they have higher costs and weaker integration with the Microsoft environment. Moreover, Defender for Endpoint is included with Microsoft 365 Business Premium and can be managed through Intune, offering a cost-effective, all-in-one solution. Also, Defender for IoT is licensed based on network traffic monitored rather than device count, allowing for budgetary flexibility for small to medium municipalities.

**Table 2:** *Cost and Alternative Tool Comparison*

| Product | Strengths | Weaknesses | Cost |
|---|---|---|---|
| **Defender** | Unified IT/OT, Microsoft stack | Best for Microsoft environments | **Included in Microsoft 365 E5 ($57/user/month)** or licensed per OT traffic volume |
| **CrowdStrike** | Fast EDR, good intel | Limited OT, expensive | $60–$100+/endpoint |
| **SentinelOne** | Easy deployment | Not OT-specific | $70–$120+/endpoint |
| **Claroty/Nozomi** | OT native, deep protocol | High cost | $100K+ per year |

**Evidence-Based Argumentation**

Academic literature and industry reports highlight growing threats to critical infrastructure. Khan, Rehman, and Abbas (2022) reviewed vulnerabilities in SCADA systems, confirming how legacy devices are often exposed due to default credentials or poor network segmentation. Sodhro et al. (2023) argue that combined IT/OT defense strategies are essential for critical infrastructure flexibility. In the case of the Aliquippa cyberattack, their basic endpoint, the PLC, was the path for the attack due to misconfiguration. Microsoft Defender could block this path with alerts for remote access attempts, default passwords, and unauthorized actions, all flagged through known attacker tactics.

To add on, Microsoft Defender's built-in altering, investigation, and automated containment features directly support this lifecycle mentioned in Chapter 4 on communication and network security, emphasizing the incident response lifecycle, illustrating how proper identification, containment, eradication, recovery actions can minimize breach impacts (Chapple, Stewart, & Gibson, 2024). And lastly, Chapter 8 highlights the importance of security assessment and testing, emphasizing that vulnerabilities must be proactively identified and mitigated. Defender tools enable regular scanning, real-time monitoring, and vulnerability assessments, fulfilling these needs.

**Installation and Risk Evaluation**

Deployment of Defender for Endpoint can occur in two different ways, through Intune or manual installation on control PCs. Defender for IoT uses passive network monitoring to detect abnormal behavior on OT devices without disrupting operations. However, this deployment could cause staff unfamiliarity, false positives, and compatibility with legacy systems. These can be mitigated through gradual deployment, staff training, and enabling automatic containment features. Integration with Microsoft Secure Score also helps quantify risk reduction over time.

We can apply the security operation principles of Chapter 9 at this stage, especially regarding centralized monitoring, alerting, and incident management (Chapple, Stewart, & Gibson, 2024). Additionally, Chapter 15 introduces business continuity and disaster recovery planning, which underscores the need for continuity strategies. Defender ensures this by enabling real-time containment and providing post-breach diagnostics, which maintains essential operations.

**Table 3:** *Installation Phase for Defender Deployment*

| Phase | Description |
| --- | --- |
| **Phase 1** | Review network setup and identify vulnerable computers and PLCs |
| **Phase 2** | Install Defender for Endpoint on pilot workstations via Microsoft Intune |
| **Phase 3** | Set up Defender for IoT, mirror traffic with SPAN/TAP, and discover devices |
| **Phase 4** | Tune alert rules and monitor results for false positives |

**Table 4:** *Risk Evaluation and Mitigation Strategies*

| Feature | Mitigation Strategy |
| --- | --- |
| **Staff unfamiliarity** | Provide training on using Defender dashboards and interpreting alerts |
| **False positives** | Start with limited alert scope and fine-tune rules |
| **Legacy system compatibility** | Use passive monitoring instead of agent-based tools |
| **Operational disruption** | Roll out in phases to avoid interrupting live systems |

## Recommendations

It is recommended that MWAA and similar organizations immediately deploy Microsoft Defender for Endpoint on IT and engineering systems and integrate Defender for IoT to monitor PLC traffic. This solution offers real-time detection, containment, and post-incident forensic analysis. It is cost-effective, scalable, and fits well within Microsoft-heavy environments like those typically used by small municipal systems.

Adhering to Chapter 5's principle of identity and access management, organizations should enforce the core principles of cybersecurity and proper authentication mechanisms to restrict access to PLCs and administrative consoles (Chapple, Stewart, & Gibson, 2024). From Chapter 17, which discusses secure operations architecture, Defender provides system hardening, attack surface reduction, and centralized security management, meeting the components of a secure system lifecycle. Finally, Chapter 18 on software development security supports using trusted, regularly updated security tools like Microsoft Defender, which satisfy the secure coding standards and routine patch management.

**Table 5:** *Benefits of Microsoft Defender for Endpoint*

| Feature | What It Does |
|---|---|
| **Watches behavior** | Looks for unusual or dangerous activity |
| **Blocks threats** | Automatically stops malware and isolates devices |
| **Finds weak spots** | Checks for outdated software or common weaknesses |
| **Works with Microsoft 365** | Combines alerts from different Microsoft tools |
| **Controls access** | Uses secure logins and user permissions |

**Table 6:** *Benefits of Microsoft Defender for IoT*

| Feature | What It Does |
|---|---|
| **Monitors OT traffic** | Watches device communication without interrupting them |
| **Finds all devices** | Lists all OT devices on the network |
| **Inspects protocols** | Detects strange commands to devices like PLCs |
| **Checks configurations** | Flags devices using default settings or open ports |
| **Supports segmentation** | Detects when devices cross into restricted network areas |

**Conclusion**

The Aliquippa breach revealed how legacy devices and limited visibility can open the door to critical infrastructure cyberattacks. With cyberattacks on utilities increasing and over 70% of systems failing to meet cybersecurity standards (WESA, 2024), tools like Microsoft Defender offer a framework-based, cost-effective response. Defender for Endpoint and Defender for IoT work together to identify, contain, and prevent threats before damage occurs.

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